

**ECON 101—Introduction to Microeconomics**  
**Fall 2019**

**Midterm I-Retake**

October 11, 2019

**Multiple choice**

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|    |    |    |    |    |    |    |    |    |    |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|    |    |    |    |    |    |    |    |    |    |

**Open problems**

Problem 1:

1. The coordinate of point B is \_\_\_\_\_ and the coordinate of point C is \_\_\_\_\_.
2. The Income effect is \_\_\_\_\_ and the substitution effect is \_\_\_\_\_.
3. The cross price elasticity is \_\_\_\_\_.

Problem 2:

1. The aggregate demand is: \_\_\_\_\_.
2. The Firm's supply schedule is: \_\_\_\_\_.
3. The equilibrium price is \_\_\_\_\_ and the equilibrium quantity is \_\_\_\_\_.
4. Firms will \_\_\_\_\_.
5. The long run equilibrium price is \_\_\_\_\_, the long run equilibrium quantity is \_\_\_\_\_ and the number of firms is \_\_\_\_\_.

Part I: Multiple Choice

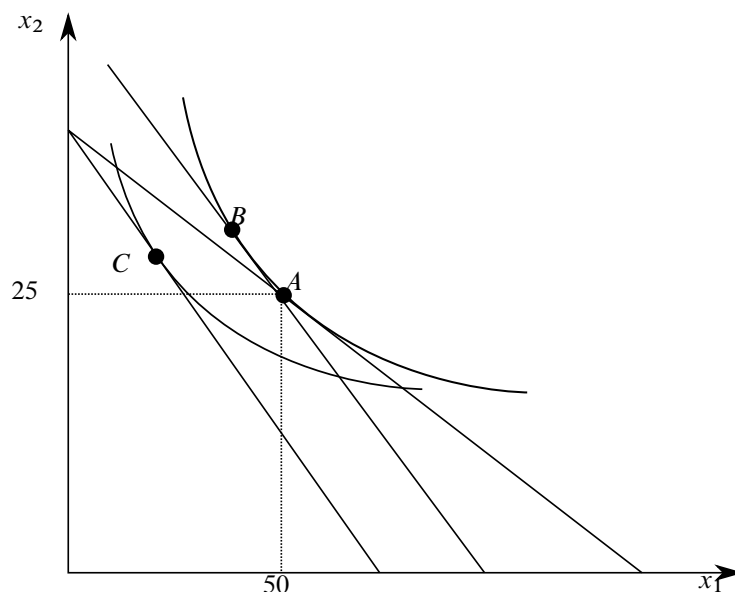
Please record all questions in the cover sheet. Each question is worth 2 points for a total of 40 points.

| Version  | I   | II  | III | IV  | V   |
|----------|-----|-----|-----|-----|-----|
| Question |     |     |     |     |     |
| 1        | (a) | (b) | (d) | (b) | (d) |
| 2        | (d) | (d) | (c) | (a) | (d) |
| 3        | (c) | (b) | (b) | (b) | (a) |
| 4        | (b) | (b) | (a) | (a) | (d) |
| 5        | (a) | (b) | (d) | (d) | (b) |
| 6        | (b) | (a) | (a) | (b) | (b) |
| 7        | (a) | (b) | (d) | (d) | (a) |
| 8        | (c) | (a) | (a) | (c) | (a) |
| 9        | (d) | (b) | (a) | (c) | (b) |
| 10       | (c) | (c) | (b) | (b) | (c) |
| 11       | (a) | (b) | (b) | (b) | (d) |
| 12       | (a) | (a) | (b) | (c) | (a) |
| 13       | (b) | (c) | (c) | (d) | (b) |
| 14       | (a) | (c) | (c) | (d) | (d) |
| 15       | (c) | (b) | (a) | (d) | (b) |
| 16       | (c) | (a) | (a) | (d) | (a) |
| 17       | (a) | (a) | (b) | (c) | (c) |
| 18       | (c) | (d) | (c) | (c) | (a) |
| 19       | (c) | (b) | (b) | (a) | (a) |
| 20       | (c) | (a) | (a) | (b) | (d) |

## Part II: Open problems

**Instructions:** Please do your calculations here but write your final answers in the space provided in the cover sheet.

1. Consider the following scenario as depicted by the following figure.



This consumer can consume goods  $x_1$  and  $x_2$ . The consumer has an income of \$100 dollars. The price of  $x_1$  is \$1 dollar and the price of  $x_2$  is \$2. Each consumer initially chooses the initial consumption bundle  $x_1 = 50$ ,  $x_2 = 25$  which is denoted by point A. Suppose that there is an increase in the price of  $x_1$  to \$2. After this price increase, the optimal consumption bundle of the consumer is denoted by point C. Suppose the government gives the consumer a monetary transfer of \$60 so that she can achieve her original indifference curve before the price change. Also, assume that an economic researcher in the bureau of taxation gives you the following information:

- The arc own price elasticity for good 1 of the consumer is  $|\epsilon_{x_1}^{p_1}| = 1.29$ .
- The arc income elasticity with respect to good 1 of the consumer is  $\epsilon_{x_1}^I = 1.44$

With the previous information please compute (round your answers to the nearest hundredth):

- The coordinates of points B and C.
- The Income and Substitution effects.
- The (arc) cross price elasticity of good  $x_2$

Write your procedure for finding the solution here.

2. Suppose that an economy has two individuals, A and B. A's demand curve is given by  $q_D^A = 100,000 - 2p$  and B's demand curve is given by  $q_D^B = 150,000 - 6P$ . Furthermore, there are initially 10,000 identical firms with the following cost structure: Total Cost:  $TC(q) = 10,000 + 100q + 10q^2$ , Marginal Cost:  $MC(q) = 100 + 20q$ .

- (a) Find the market demand curve. Provide an expression for  $Q_D$
- (b) What is the firm's shutdown point and what is the firm's supply schedule?
- (c) What is the market price and quantity?
- (d) Will firms enter or exit?
- (e) Calculate the long-run equilibrium price, quantity, and number of firms.

Write your procedure for finding the solution here. Round your answers to the nearest hundredth.